

Business Mathematics and Statistics

Course: ATD — Accounting Technicians Diploma

Level: Level II

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Syllabus version: Based on the standard KASNEB ATD Level II Business Mathematics and Statistics syllabus topics — basic algebra, percentages and ratios, simple and compound interest, and elementary statistics. Authored from general quantitative-methods knowledge and standard syllabus topic coverage, not copied from any existing notes provider or study guide. Cross-check terminology and emphasis against the current KASNEB syllabus and examiner's reports before the exam.

Original work notice. Every explanation, worked example, and question in this document was written from scratch for this paper. No text has been copied or paraphrased from any KASNEB past paper, textbook, or third-party notes/study guide. The scenarios and practice questions are original.

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1. Basic algebra for business problems

Many business problems can be expressed as a simple linear equation and solved by isolating the unknown variable. For example, if total cost = fixed cost + (variable cost per unit × number of units), and you know total cost, fixed cost and variable cost per unit, you can rearrange to solve for the number of units:

$$\text{Number of units} = (\text{Total cost} - \text{Fixed cost}) \div \text{Variable cost per unit}$$

The key skill is translating a word problem into a correct equation before attempting to solve it.

2. Percentages, ratios and proportions

- **Percentage** — a fraction expressed out of 100; $\text{Percentage} = (\text{Part} \div \text{Whole}) \times 100$.
 - **Ratio** — compares two or more quantities directly (e.g. a 2:3 ratio of capital contributed by two partners).
 - **Proportion** — a statement that two ratios are equal, used to scale quantities up or down consistently (e.g. if 5 workers produce 200 units, how many units would 8 workers produce, at the same rate?).
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3. Simple interest

Simple interest is calculated only on the original principal amount, for every period:

$$\text{Simple Interest (I)} = \text{Principal (P)} \times \text{Rate (R, as a decimal)} \times \text{Time (T, in years)}$$

$$\text{Total amount repayable} = \text{Principal} + \text{Simple Interest}$$

4. Compound interest

Compound interest is calculated on the principal plus any interest already added in previous periods — "interest on interest":

$$\text{Amount (A)} = \text{Principal (P)} \times (1 + r)^n$$

where **r** is the interest rate per period (as a decimal) and **n** is the number of periods.

Compound interest grows faster than simple interest over time, because each period's interest is calculated on a progressively larger base.

5. Measures of central tendency — mean, median, mode

- **Mean (average)** — sum of all values $\div$ number of values; sensitive to unusually large or small values (outliers).
 - **Median** — the middle value when data is arranged in order (or the average of the two middle values, for an even number of data points); not affected by outliers.
 - **Mode** — the value that occurs most frequently in a data set; a data set can have one mode, more than one, or none at all.
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6. Measures of dispersion — range and standard deviation

- **Range** — the difference between the highest and lowest values in a data set; simple to compute, but only uses two values and ignores everything in between.
 - **Standard deviation** — measures how spread out data values are around the mean; a small standard deviation means values cluster closely around the mean, while a large one means values are widely spread. Computed, at a basic level, as the square root of the average of the squared deviations of each value from the mean.
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7. Simple index numbers

An **index number** measures the relative change in a variable (e.g. price, quantity) over time compared to a fixed base period, which is set to 100.

$$\text{Price index} = (\text{Price in current period} \div \text{Price in base period}) \times 100$$

An index of 115, for instance, means the value has risen by 15% compared to the base period.

8. Basic probability concepts

Probability measures the likelihood of an event occurring, expressed as a number between 0 (impossible) and 1 (certain).

$$\text{Probability of an event} = \text{Number of favourable outcomes} \div \text{Total number of possible outcomes}$$

For two mutually exclusive events (they cannot both happen at the same time), the probability that either occurs is the sum of their individual probabilities. For two independent events (one does not affect the other), the probability that both occur is the product of their individual probabilities.

9. Full worked question — compound interest and mean calculation

Scenario. (a) Kamau invests KSh 100,000 at 10% per year compound interest for 3 years. Compute the amount at the end of the 3 years. (b) A shop recorded daily sales (KSh '000) of 20, 25, 18, 30, 22 over five days. Compute the mean daily sales.

Solution:

$$(a) A = P(1 + r)^n = 100,000 \times (1.10)^3 = 100,000 \times 1.331 = \text{KSh } 133,100$$

$$(b) \text{Mean} = (20 + 25 + 18 + 30 + 22) \div 5 = 115 \div 5 = \text{KSh } 23,000$$

10. Practice examination — KASNEB-style paper (original, not a past paper)

Note on format. This section is laid out the way a KASNEB ATD examination paper is laid out — a timed paper with five compulsory 20-mark questions — purely so you can practise under realistic exam conditions. Every question below is an original scenario written for this document. **It is not a reproduction of any actual KASNEB past examination paper.**

ATD LEVEL II — BUSINESS MATHEMATICS AND STATISTICS (PRACTICE PAPER)

TIME ALLOWED: 3 HOURS

Instructions to candidates: Answer ALL FIVE questions. Each question carries 20 marks. Show all workings clearly.

QUESTION ONE

A trader's total cost is given by: Total cost = 40,000 + (250 × number of units). Last month, total cost was KSh 115,000.

- (a) Form the equation and solve for the number of units produced. (8 marks)
- (b) A quantity increases from 400 to 460. Compute the percentage increase. (6 marks)
- (c) Two partners contribute capital in the ratio 3:5. If total capital is KSh 320,000, compute each partner's contribution. (6 marks)

(Total: 20 marks)

QUESTION TWO

- (a) Compute the simple interest on KSh 250,000 at 8% per year for 4 years, and the total amount repayable. (8 marks)
- (b) Compute the compound amount on KSh 150,000 at 12% per year, compounded annually, for 2 years. (8 marks)
- (c) Briefly explain why compound interest grows faster than simple interest over time. (4 marks)
- (Total: 20 marks)
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QUESTION THREE

The weekly wages (KSh '000) of seven employees are: 12, 15, 12, 18, 20, 12, 16.

- (a) Compute the mean wage. (5 marks)
- (b) Compute the median wage. (5 marks)
- (c) Identify the mode. (4 marks)
- (d) Compute the range of the wages. (6 marks)
- (Total: 20 marks)
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QUESTION FOUR

The price of a commodity was KSh 40 in the base year and is KSh 52 in the current year.

- (a) Compute the price index for the current year. (6 marks)
- (b) Interpret the index number you have computed. (4 marks)
- (c) A bag contains 4 red balls and 6 blue balls. One ball is drawn at random. Compute the probability that it is (i) red, (ii) blue. (10 marks)
- (Total: 20 marks)
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QUESTION FIVE

- (a) Distinguish between the mean, median and mode as measures of central tendency, stating one situation where the median is preferred over the mean. (12 marks)
- (b) Explain what the standard deviation measures, and what a small standard deviation indicates about a data set. (8 marks)
- (Total: 20 marks)
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11. Marking guide — model answers to the practice examination

QUESTION ONE

- (a) (8 marks) — $115,000 = 40,000 + 250x \rightarrow 250x = 75,000 \rightarrow x = \mathbf{300 \text{ units}}$.
- (b) (6 marks) — Increase = 60; Percentage increase = $(60 \div 400) \times 100 = \mathbf{15\%}$.
- (c) (6 marks) — Total ratio parts = 8; Partner A = $(3/8) \times 320,000 = \mathbf{KSh 120,000}$; Partner B = $(5/8) \times 320,000 = \mathbf{KSh 200,000}$.

QUESTION TWO

- (a) (8 marks) — $I = 250,000 \times 0.08 \times 4 = \mathbf{KSh 80,000}$; Total repayable = $250,000 + 80,000 = \mathbf{KSh 330,000}$.
- (b) (8 marks) — $A = 150,000 \times (1.12)^2 = 150,000 \times 1.2544 = \mathbf{KSh 188,160}$.
- (c) (4 marks) — Because compound interest is calculated on the principal plus previously added interest each period, so the base on which interest is calculated grows each period, unlike simple interest which stays on the original principal only.

QUESTION THREE

- (a) (5 marks) — Mean = $(12+15+12+18+20+12+16) \div 7 = 105 \div 7 = \mathbf{KSh 15,000}$.
- (b) (5 marks) — Ordered: 12,12,12,15,16,18,20 $\rightarrow$ Median (4th value) = $\mathbf{KSh 15,000}$.
- (c) (4 marks) — Mode = $\mathbf{KSh 12,000}$ (occurs three times).
- (d) (6 marks) — Range = $20 - 12 = \mathbf{KSh 8,000}$.

QUESTION FOUR

- (a) (6 marks) — Price index = $(52 \div 40) \times 100 = \mathbf{130}$.
- (b) (4 marks) — The price has risen by 30% compared to the base year.
- (c) (10 marks) — Total balls = 10. (i) $P(\text{red}) = 4/10 = \mathbf{0.4}$. (ii) $P(\text{blue}) = 6/10 = \mathbf{0.6}$.

QUESTION FIVE

- (a) (12 marks) — Mean: sum of values $\div$ number of values, sensitive to outliers. Median: the middle value when ordered, unaffected by outliers. Mode: the most frequently occurring value. The median is preferred over the mean when data contains extreme outliers that would distort the mean (e.g. one very high salary in an otherwise modest wage list).
- (b) (8 marks) — Standard deviation measures how spread out data values are around the mean; a small standard deviation indicates values cluster closely around the mean (low variability), while a large one indicates values are widely spread (high variability).

12. Exam technique and revision checklist

- Word-problem questions: always write the equation explicitly before solving — marks are typically awarded for correctly forming the equation, not just the final number.
- Interest questions: state whether the question is simple or compound interest clearly, and use the correct formula ($I = PRT$ for simple; $A = P(1+r)^n$ for compound) — don't mix the two up.
- Central tendency questions: always order the data first before finding the median, and double-check for ties when identifying the mode.
- Index number questions: always state the formula ($\text{Current} \div \text{Base} \times 100$) and interpret the resulting number in plain words when asked to "interpret."
- Probability questions: state the total number of possible outcomes clearly before computing each individual probability.

Quick revision — one line per topic:

1. Percentage = $(\text{Part} \div \text{Whole}) \times 100$.
2. Simple interest: $I = P \times R \times T$.
3. Compound interest: $A = P(1 + r)^n$.
4. Mean = $\text{sum} \div \text{count}$; Median = middle value (ordered data); Mode = most frequent value.
5. Range = highest – lowest value.
6. Standard deviation: measures spread around the mean; smaller = less variability.
7. Price index = $(\text{Current price} \div \text{Base price}) \times 100$.
8. Probability = favourable outcomes $\div$ total possible outcomes.

End of study notes. This document is intended as original exam-preparation material and should be used alongside the current official KASNEB syllabus and examiner's reports for the paper.